ABDULLAH SHAMAIL

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PROFESSIONAL SUMMARY

PhD Candidate specializing in Spatio-temporal Data Analysis and Spatio-temporal Statistical Machine Learning. Proven track record in developing scalable discovery algorithm and predictive architectures (GNNs, Transformers) for complex dynamic systems.

EDUCATION

08/2026 **DOCTOR OF PHILOSOPHY, COMPUTER ENGINEERING** *(Expected)*
Iowa State University (ISU), Ames, IA
Major Professors: Goce Trajcevski, Ashfaq Khokhar
Thesis: Constraint-Aware Scalable Discovery of Persistent Spatio-Temporal Patterns

12/2024 **MASTER OF SCIENCE, COMPUTER ENGINEERING**
Iowa State University (ISU), Ames, IA
Major Professors: Goce Trajcevski
Specialization: Software Systems

06/2020 **BACHELOR OF SCIENCE, COMPUTER SCIENCE**
Lahore University of Management Sciences (LUMS), Lahore, Pakistan
Major Professor: Muhammad Fareed Zaffar
BS Thesis: Authorship Attribution and Obfuscation

RESEARCH EXPERIENCE

06/2022 – Present **GRADUATE RESEARCH ASSISTANT – ISU**

SDC Framework (Submitted - 2026)

Designed the Semantic Diversity in Convoys (SDC) framework to advance diversity-aware clustering in complex dynamic environments (e.g., molecular dynamics, biological movement).

Implemented Brute-Force, ED-SDC, and Grid-SDC algorithms, leveraging indexing and spatial partitioning to achieve 17x to 65x computational speedups.

Integrated transformers and embedding-based similarity measures to efficiently process and identify persistent spatio-temporal patterns.

Motif Mining and Scientific ML (Pending Submission - 2026)

Developing large-scale motif-mining pipelines leveraging geometric descriptors, spatio-temporal statistics, and learned embeddings to identify aromatic ring-pair configurations predictive of HB persistence - enabling new scientific ML approaches for event prediction and simulation understanding.

Bond-Aware Relaxed Moving Clusters (BARMC) (SIGSPATIAL 2024, TSAS 2026)
Introduced, developed, and implemented the Bond-Aware Relaxed Moving Clusters (BARMC) algorithm.
Optimized the detection of key interactions within complex datasets by explicitly integrating spatial proximity metrics with domain-specific semantic rules.
Enhanced traditional clustering techniques, successfully achieving a ~1.7x reduction in computation time for complex trajectory data.

Automatic Detection of Hydrogen Bonds (HBs) (SIGSPATIAL 2024)
Designed and implemented a software system to perform real-time detection and analysis of dynamic formations and their persistence in simulations.
Automated the identification of long-lasting interactions and generated detailed analytical reports, reducing computational overhead for complex simulations.
Developed robust visualization components for dynamic interactions to streamline the overall data analysis process.

Compressing Molecular Dynamics (MD) Data (ADBIS 2022)
Developed domain-specific trajectory compression algorithm for MD simulations, achieving an 80% reduction in storage requirements without sacrificing query effectiveness.
Implemented the algorithm to optimize within-distance query processing, achieving a 3x speedup in data retrieval for complex simulations. This led to the publication of an award-winning paper in Advances in Databases and Information Systems (ADBIS 2022).

01/2021 – 05/2022 **INDEPENDENT RESEARCHER – ISU**

Large-Scale Data Analysis: Leveraged Java and Hadoop MapReduce API for large-scale data analysis, focusing on text analytics. Developed and optimized an algorithm to efficiently identify the top 10 most frequent bigrams in extensive text corpora, achieving a 2x speedup in data processing. Also engineered a Word Count algorithm on Shakespeare and Gutenberg datasets, confirming the program's scalability and robustness.

Sarcasm Detection in News Headlines: Leveraged Python and advanced machine learning algorithms, including LSTM, SVM and Decision Trees, to develop a sarcasm detection model for news headlines sourced from a Kaggle dataset. Achieved a high classification precision of 84% and recall of 82**%**, employing dense vector representations for tokenized and padded input, improving the computer's ability to identify sarcasm in textual data.

Deep Convolutional GAN for High-Quality Emoji Generation: Successfully implemented and trained a Deep Convolutional Generative Adversarial Network (DCGAN) for emoji generation, achieving optimal performance metrics with a Discriminator Loss of 0.426 and a Generator Loss of 1.665 after 2500 epochs. Applied research-backed stabilization techniques to further refine the model, that generated emojis that closely resembled real-world examples, demonstrating its robustness and effectiveness.

Classification using Layer-Optimized Convolutional Neural Networks: Executed a two-phase project to classify the CIFAR-10 dataset using Convolutional Neural Networks (CNNs). In the first phase, manually implemented a CNN, overcoming the inherent challenges of image classification. In the second phase, utilized Keras to experiment with advanced layers including Pooling, Dropout, and Batch Normalization. Conducted a model comparison, achieving a high classification accuracy of 72% with an optimized, multi-layered CNN model.

TEACHING EXPERIENCE

01/2021 – Present **TEACHING ASSISTANT– ISU**

Introduction to Computer Engineering and Problem Solving (Fall 2025)
Delivered foundational instruction to 150+ first-year students, introducing core computer engineering principles through project-based learning like algorithmic problem solving, C/C++ programming, and technical report writing. Guided small and large student groups through the full lifecycle of computer-based projects. Focused on developing "interactive skills" and professional presentation standards, helping students transition from high school studies to rigorous engineering problem-solving methodologies. Ensured timely and detailed feedback to students, helping them understand their progress well.

Software Evolution and Maintenance (Spring 2025)
Facilitated the study of long-term software lifecycles, emphasizing systematic defect analysis and debugging. Mentored undergraduate senior students in tracing and understanding large, complex codebases. Evaluated professional-grade written reports and oral presentations, providing feedback on technical communication and experimental studies of software maintenance.

Digital Logic Circuits (Fall 2024)
Supervised laboratory sessions, by introducing content with a mini lecture each time, for 25+ undergraduate sophomore/junior students, on Boolean algebra, Finite State Machines (FSMs), Verilog/VHDL hardware description languages, and programmable logic devices (FPGA). Instructed students on computer-aided schematic capture and simulation tools. Provided hands-on guidance for building arithmetic circuits and sequential logic, ensuring

students could translate abstract logic minimization into functional physical hardware.

Software Tools for Large-Scale Data Analysis (Spring 2024, Spring 2025)
Led weekly laboratory sessions, by introducing content with a mini lecture each time, for 30+ graduate and/or undergraduate students, on external memory processing, stream processing, large-scale parallelism (Hadoop/Spark/Pig Latin), and data interchange formats. Bridged the gap between theoretical data science and practical software engineering by teaching students how to manage high-volume data within distributed environments. Ensured timely delivery of all grades and feedback for all students to help them understand their errors.

Software Analysis & Verification for Safety & Security (Fall 2023)
Guided 80+ graduate and/or undergraduate senior students through the mathematical foundations of software safety, threat modeling, and vulnerability categorization for cyber-physical systems by holding regular office hours and tutorials. Facilitated case studies on large-scale systems software, helping students implement verification algorithms and navigate the evolving landscape of security mitigation techniques. Provided technical mentorship on program comprehension and the use of professional analysis tools. Graded all course content for the students, ensuring timely delivery of feedback and grades.

Introduction to Spreadsheets and Databases (Fall 2021, Spring 2021)
Managed the instructional logistics for a high-enrollment foundational course, with over 200 undergraduate (freshmen-sophomore) students enrolled throughout each of the semesters, achieving high efficiency through the development of automated grading scripts. Delivered lab lectures on using Microsoft Excel and Access and guided students with small tutorials on each of the topics being discussed. To support the diverse and large cohort, held regular office hours and developed automated grading scripts to provide students with rapid feedback on their assignments and homework. Focused on making myself accessible and available, making sure students did not face any hurdles while learning.

08/2018 – 05/2020 **UNDERGRADUATE TEACHING ASSISTANT – LUMS**

Deep Learning (Spring 2020)
Delivered guest lectures and led laboratory sessions focused on the implementation of neural network architectures, for both graduate and undergraduate students enrolled. Simplified complex mathematical concepts such as backpropagation, stochastic gradient descent, and loss function optimization. Designed and graded the assignments that covered all the mentioned topics. Conducted specialized tutorials on hyperparameter tuning and model evaluation metrics to ensure student success in final projects.

Discrete Mathematics (Fall 2019)
Held tutorials for undergraduate students on the mathematical foundations of computer science, including Set Theory, Logic, Combinatorics, and Graph Theory. I held dedicated recitation sessions to help students master the art of formal proofs and induction. Developed practice problem sets and exam materials that emphasized the application of discrete math in algorithm design and data structures.

Computational Problem Solving (Fall 2018, Spring 2019)
Managed instructional logistics for a high-enrollment foundational course. Led weekly recitations and extra tutorials designed to help a diverse student body master algorithmic thinking and syntax. Assisted in the creation of interactive class activities that broke down complex engineering problems into manageable logical steps. Focused on teaching students how to move from a conceptual flowchart to a functional implementation in C++. Graded exams and programming assignments for 200+ students, providing detailed feedback to bridge the gap between theoretical logic and practical debugging. Implemented "early alert" tutorials for students struggling with core programming constructs, significantly improving overall class performance.

INDUSTRY EXPERIENCE

05/2025 – 08/2025 **ENGINEERING DATA SCIENCE INTERN,** Altair Engineering, Troy, MI

Developed and optimized GNN, Transformer, and RNN-based architectures for Altair's PhysicsAI models, analyzing large-scale spatio-temporal physical simulation data.

Achieved a 10x reduction in Mean Absolute Error (MAE), enabling higher-fidelity and longer-horizon predictions for complex physical systems.

Built and deployed end-to-end PyTorch and TensorFlow pipelines supporting training, evaluation, and GPU-accelerated inference within a production simulation platform.

Engineered real-world case studies from these GPU-accelerated pipelines, which are now used to teach scalable spatio-temporal modeling to senior students.

05/2018 – 06/2018 **SOFTWARE DEVELOPER,** Taar Consulting, Lahore, Pakistan

Designed and implemented a Customer Feedback Plugin for Retail Pro Prism, leveraging its APIs after thorough mastery. The plugin was seamlessly integrated into existing transaction systems, capturing feedback from **over 1,000 daily transactions**. This system provided actionable insights to enhance the customer experience and improve business operations.

MENTORING EXPERIENCE

01/2022 – 05/2023 **PROJECT MENTORING – ISU (SPRING/FALL 2022, SPRING 2023)**

Mentored multiple senior-year capstone teams (100+ students) through the full Software Development Life Cycle. Guided students in making critical architectural decisions, selecting appropriate technology stacks, and implementing Agile project management methodologies. Facilitated weekly "sprint" reviews to resolve technical roadblocks, conduct code reviews, and ensure students adhered to industry standards for software quality, security, and documentation. Successfully mentored a student group in refining their project for academic publication. This led to a co-authored demo paper at ACM SIGSPATIAL 2024, showcasing my ability to elevate undergraduate coursework to the level of internationally recognized research.

PUBLICATIONS

PEER-REVIEWED JOURNAL PUBLICATIONS

1. Hasan Anowar, M., **Shamail, A.**, Wang, X., Trajcevski, G., Murad, S., Jameson, C. J., & Khokhar, A. (2024). Compressing generalized trajectories of molecular motion for efficient detection of chemical interactions. **Information Systems**, 125, 102426. https://doi.org/10.1016/j.is.2024.102426
2. Mubin, O., Alnajjar, F., **Shamail, A**. *et al.* The new norm: Computer Science conferences respond to COVID-19. ***Scientometrics*** 126, 1813–1827 (2021). https://doi.org/10.1007/s11192-020-03788-9

CONFERENCE PROCEEDINGS

1. **Shamail, A** et al. Bond-Aware Moving Clusters of Atomic Trajectories with Relaxed Persistency. In Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '24). Association for Computing Machinery, New York, NY, USA, 605–608. https://doi.org/10.1145/3678717.3691298

2. Anowar, M.H, **Shamail, A** *et al.* Detecting and Visualizing Bond-Forming Convoys in Atomic and Molecular Trajectories. In Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '24). Association for Computing Machinery, New York, NY, USA, 657–660. https://doi.org/10.1145/3678717.3691264

3. Anowar, M.H, **Shamail, A** *et al.* (2022). Generalization Aware Compression of Molecular Trajectories. In: Chiusano, S., Cerquitelli, T., Wrembel, R. (eds) Advances in Databases and Information Systems. **ADBIS 2022**. Lecture Notes in Computer Science, vol 13389. Springer, Cham. https://doi.org/10.1007/978-3-031-15740-0_20

JUST ACCEPTED

1. **Shamail et al.** 2026. Bond-Aware Moving Clusters of Atomic Trajectories with Relaxed Persistency. ACM Trans. Spatial Algorithms Syst. Just Accepted (March 2026). https://doi.org/10.1145/3804446
2. **Shamail et al.** 2026 et al. Semantically Diverse Convoys (Accepted to MDM '26)

SERVICE AND LEADERSHIP

SERVICE TO DISCIPLINE

11/2024 **Volunteer, ACM SIGSPATIAL 2024 – Atlanta, Georgia**

Contributed to the successful execution of a top-tier international conference by managing the registration desk, coordinating logistics for 300+ attendees, and facilitating technical sessions. Helped with the setup of research poster sessions, ensuring optimal flow and visibility for presenters. Acted as a point of contact for attendees, providing venue guidance and technical assistance during plenary and break-out sessions.

SERVICE TO CAMPUS

05/2023 – 05/2024 **President - Pakistan Student Association (PSA) – ISU**

Led community engagement initiatives, notably organizing a Pakistan Independence Day event that attracted over 80 attendees. Oversaw budget management and financial planning, utilizing data-driven approaches to optimize resource allocation. Designed and executed a social media campaign, leveraging real-time data analytics to refine strategies. This ultimately helped raise $20,000 for flood relief efforts, which supported 15 million affected individuals in Pakistan. Participated in, and cooked for, the International Culture and Food Fest attended by more than 300 attendees at ISU. Volunteered weekly at ISU Dining services to help them cater to the dining demands of the students.

05/2019 – 05/2020 **Vice President - Index: Design and Innovation Society – LUMS**

Initiated the first-ever Design Innovation Event, using quantitative metrics to identify key areas of interest and measure outcomes. Mentored a cohort of 50 students, employing performance indicators and KPIs to track progress and assess impact. Successfully organized UX Pakistan, an international design conference that attracted approximately 500 global attendees annually from 2018 to 2020. Additionally, hosted the National Design Awards in Pakistan in 2018.

HONORS AND AWARDS

1. Recipient of the **NSF (National Science Foundation) Student Travel Grant**, awarded through a competitive selection process to support conference attendance and professional development.
2. Best Paper award for the research: Generalization Aware Compression of Molecular Trajectories at **ADBIS 2022**
3. Awarded the title of **"Preparing Future Faculty Associate"** by Iowa State University in 2025
4. **Captain** of High School Soccer Team in 2015-2016
5. **President of the Mathematics Society** in High School in 2015-2016

SPOKEN LANGUAGES

English (fluent), Urdu (native), Punjabi (intermediate)

SOFTWARE AND CODING

Data Analysis and Machine Learning: TensorFlow, PyTorch, Sklearn, Keras, Pandas, Numpy, Apache Spark, Apache Flink, Hadoop, Pig Latin, SQL, MATLAB, SQL Server, MDX, MySQL.

Core Languages: Python, C++, Objective C, JavaScript/ReactJS.

Familiarity: GoLang, Haskell, Java, PHP.

Miscellaneous: LaTeX, GitHub, Trajectory Analysis, Slack, Discord, MS Excel, MS Access.

REFERENCES

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